

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Photovoltaic Templates

WECC Dist. Small PV Plants 25MVA 50Hz,
WECC Dist. Small PV Plants 25MVA 60Hz,
WECC Large-scale PV Plant 110MVA 50Hz,
WECC Large-scale PV Plant 110MVA 60 Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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Contents

1 Introduction **1**

2 Model Usage **2**

3 Model Configuration **2**

3.1 Adjusting Rated Power 2

3.2 Adjusting Power Dispatch 2

3.3 Dynamic Model Configuration 2

4 Model Documentation **3**

4.1 Large-scale PV Plant 3

4.2 Distributed PV Plant 4

5 References **5**

List of Tables **6**

1 Introduction

This document describes the usage of the WECC PV Power Plant models implemented in *PowerFactory*. The models are based on [1, 8].

The following plant model types defined by WECC are available in *PowerFactory*:

- Large-scale PV plants
- Distributed and small PV plants (PVD1, superseded by DER_A, see note below)

Chapters 2 and 3 describe how to use and configure the models in *PowerFactory*. In chapter 4 the different models are described.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. The user has the task to adjust static and dynamic model parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific PV plant!

Note: The PVD1 model has been superseded by the DER_A model as suggested by WECC REMTF [9]. The DER_A model is seen as an improvement over the PVD1 model (see Section “Background and Overview of the DER_A Model” of [6]). *DIgSILENT* therefore recommends to use the “WECC DER Generation” templates from the *DIgSILENT* Library instead of the “WECC Dist. Small PV Plants” templates.

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2 Model Usage

Each of the types consist of at least two grid elements, the PV system (*ElmPvsys*), a terminal and of a composite model which defines the dynamics of the PV power plant models. The composite model contains the dynamic modelled built in DSL (*DIgSILENT* Simulation Language) and all necessary measurement devices.

The models are available as templates and are accessible to be used in a project thought the button *General Templates* in the drawing toolbox.

3 Model Configuration

The rated power, dispatched power and number of parallel machines can be adjusted as explained next.

3.1 Adjusting Rated Power

The rated power of the PV system has to be configured on the *Basic Data* page of the element itself (i.e. the PV system doesn't need a type). The rated active power is calculated via the nominal apparent power "sgn" and the power factor "cosn".

3.2 Adjusting Power Dispatch

The dispatch of the PV plant can be entered on the *Load Flow* page of the PV system element. The default input mode is "P, cos(phi)", which allows to enter the operation active power and power factor. The reactive power can be limited by the reactive power capability curve which can be also voltage dependent. It is also possible to coordinate the reactive power output of several PV systems by using a station controller (*ElmStactrl*).

3.3 Dynamic Model Configuration

The dynamic simulation is started from steady state, which is the load flow result. It is therefore important to check the load flow configuration of the elements as mentioned in the previous sections.

The dynamic model can either be accessed via the generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the *Basic Data* page and click on the arrow button after "Model".
- Via Data Manager: Search the corresponding composite model , right click and select "Edit".

After opening the composite model the different dynamic (common) models can be accessed via a double click on the 2nd column.

The dynamic (common) models can be configured via scalar input parameters. These parameters are displayed after opening a common model on the *General* page.

4 Model Documentation

In the following sub-sections the different types are described. In Table 4.1 the different dynamic models are listed.

Table 4.1: Block Overview

DSL Model	Block Definition	Large-scale PV Plant	Distributed PV Plant
REEC_D Electrical Control Model	REEC_D	X	
REGC_C Generator-Converter Model	REGC_C	X	
PVD1 Distributed PV Model	PVD1		X
WTGWGO_A Weak Grid Option Model	WTGWGO_A	X	
Protection Model	Protection	X	
Plant Control	Frame WECC Plant Control	(X)	

4.1 Large-scale PV Plant

The Large-scale PV plant model is represented in the network via the PV system element. The initial settings for the load flow calculation and dynamic simulation are done in the same element, configuration of the grid element is described in 3.

The model is recommended to represent aggregated plants where a single equivalent PV generator is enough to represent the total output power and dynamics, the model as build in *PowerFactory* consist of the following components:

- Large-scale PV Plant.
- REGC_C, which implements the fast converter controllers and represents the interface between the generator/converter [8].
- REEC_D, which implements the electrical controls of the inverter, calculating active and reactive current references based on power references, voltage and power measures. Different control modes are allowed depending on the flag settings as shown in Table 4.2.
- WTGWGO_A, which implements a supplemental weak grid option control [7].
- Protection, which implements the protection functions with disconnection from the grid in case of over- or under- frequency, over- or under- voltage. The protection model is in the template by default out of service.
- Plant Control, which implements the plant controller which calculates the commanded active and reactive current. The plant control is available as template in the templates folder “Plant Controllers”.
- Voltage and power measurement.

Table 4.2: Electrical control modes [3]

Control Mode	PfFlag	Vflag	Qflag
Local constant Q control	0	1	0
Local constant pf control	1	1	0
Local voltage control	0	0	1
Local coordinated Q/V control	0	1	1

Note: The REGC_C model is by default applied in the Large-scale PV templates; however, the REGC_A or REGC_B model can be used instead if desired [5]. Both, REGC_A model and REGC_B model type, are available within the DIgSILENT Library\Dynamic Models\Inverter Based Resources/Storage\WECC.
The REGC_B and REGC_C model are defined as a controlled voltage source [8]. The main difference between the REGC_B and REGC_C is that the last one includes a simplified phase-lock loop (PLL) model. The former REGC_A model was defined as a controlled current source [2].
The static generator element is configured to be a *voltage source* model in RMS simulation, i.e. to represent a controlled voltage source in accordance with REGC_C or and REGC_B.
If the REGC_A model is used, the *current source* model has to be selected in the RMS simulation settings of the static generator element. *DIgSILENT* recommends to use the additional built-in option *dq-reference angle delay* of the static generator for better numerical robustness, especially when simulating deep voltage dips. This option is described in detail in the technical reference paper of the static generator element [4]. The *delay time constant* should correspond to a typical time constant of a PLL.

4.2 Distributed PV Plant

The distributed PV plant model is represented in the network via the PV system element. This model represents a large number of PV systems connected to distribution systems symbolising the PV generation at an equivalent medium voltage feeder. The model as build in *PowerFactory* consist of the following components:

- Distributed Small PV Plants.
- PVD1. This is a simple representation of the basic control of PV generation connected to distribution system. Offering options such as volt/var control, active power controls and disconnection of generation.
- Protection. This model implements protection functions with disconnection from the grid in case of over- or under- frequency, over- or under- voltage. The protection model is in the template by default out of service.
- Voltage and power measurement.

Note: The PVD1 model has been superseded by the DER_A model as suggested by WECC REMTF [9]. The DER_A model is seen as an improvement over the PVD1 model (see Section “Background and Overview of the DER_A Model” of [6]). *DIgSILENT*therefore recommends to use the “WECC DER Generation” templates from the *DIgSILENT*Library instead of the “WECC Dist. Small PV Plants” templates.

5 References

[1] WECC - PV Power Plant Dynamic Model Specification; Issued april, 2014, 2014.

[2] WECC Second Generation Wind Turbine Models, January 2014.

[3] Model User Guide for Generic Renewable Energy System Models. Technical Report 3002006525, EPRI, June 2015.

[4] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.

[5] WECC Brief Summary of all 2nd Generation Generic Renewable Energy System Dynamic Models, February 2021.

[6] Reliability Guideline – Parameterization of the DER_A Model. Technical report, North American Electric Reliability Corporation (NERC), September 2019.

[7] P. Pourbeik. Proposal for new features for the renewable energy system generic models 07/23/18 (revised 8/3/18; 11/14/18; 11/18/18; 12/13/18; 1/20/19; 2/7/19; 3/5/19; 6/24/19; 6/28/19; 7/3/19; 7/17/19; 8/19/19; 12/17/19; 8/14/20; 8/24/20;8/31/20; 11/12/20; 4/6/21; 4/15/21; 9/22/21; 11/11/21; 9/19/22). 2022.

[8] P. Pourbeik. Proposal for new features for the renewable energy system generic models 07/23/18 (revised 8/3/18; 11/14/18; 11/18/18; 12/13/18; 1/20/19; 2/7/19; 3/5/19; 6/24/19; 6/28/19; 7/3/19; 7/17/19; 8/19/19; 12/17/19; 8/14/20; 8/24/20;8/31/20; 11/12/20; 4/6/21; 4/15/21; 9/22/21; 11/11/21; 9/19/22; 12/12/22; 1/5/23). 2023.

[9] P. Pourbeik. Proposal for DER_A Model: memo issued to WECC REMTF, 10/11/16 (revised 11/16/16; 3/6/17, 3/15/17; 3/28/17; 3/29/17; 3/31/17; 4/17/17; 10/5/17; 11/9/17; 2/9/18; 2/15/18; 3/9/18; 7/17/18; 8/29/18; 9/11/18), 2018. Technical report, WECC, September 2018.

List of Tables

4.1 Block Overview 3

4.2 Electrical control modes [3] 4